

3.1 The Wavefunction and Born Probability

In non-relativistic quantum mechanics, the physical state of a particle is completely described by a complex-valued wavefunction $\Psi(x, t)$ belonging to a Hilbert space.

According to the Born interpretation, the quantity $|\Psi(x, t)|^2$ represents the probability density of locating the particle at position x at time t . The total probability integrated across all space must equal unity.

The principle of quantum superposition establishes that if Ψ_1 and Ψ_2 are valid quantum states, any linear combination $c_1 \Psi_1 + c_2 \Psi_2$ is also a physically realizable quantum state.

3.2 The Time-Dependent Schrödinger Equation

The temporal evolution of the wavefunction is governed by the time-dependent Schrödinger equation: $i \hbar \frac{d\Psi}{dt} = H \Psi$, where H is the Hamiltonian operator representing the total energy observable.

For a non-relativistic particle of mass m in potential $V(x)$, the Hamiltonian operator takes the form $H = -(\hbar^2 / 2m) \frac{d^2}{dx^2} + V(x)$. The spatial derivative operator corresponds to the kinetic energy observable.

Stationary states arise when the potential V is time-independent, allowing separation of variables into energy eigenstates satisfying the time-independent Schrödinger eigenvalue problem $H \psi_n = E_n \psi_n$.

3.3 Heisenberg Uncertainty Principle & Observables

Physical observables in quantum theory correspond to self-adjoint Hermitian operators acting on the state Hilbert space. The eigenvalues of an observable operator correspond to the allowed measurement outcomes.

When two observable operators A and B do not commute, their commutator $[A, B] = AB - BA$ is non-zero. The Robertson-Schrödinger theorem establishes that the product of their measurement uncertainties satisfies $\sigma_A \sigma_B \geq (1/2) |\langle [A, B] \rangle|$.

For the canonical position operator x and momentum operator $p = -i \hbar d/dx$, the commutator is $[x, p] = i \hbar$, yielding the foundational Heisenberg uncertainty relation $\sigma_x \sigma_p \geq \hbar / 2$.